Label and Recover Coherent Errors

Randomized Compiling Does Not Destroy
Coherent-Error Information

Complete Mathematical Notes

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Blue equations / boxes = new results from this work

Gray equations / boxes = established results used as tools

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Setting the Scene: Quantum Circuits and Coherent Errors

The state and the circuit

A quantum processor applies a sequence of gates to an n-qubit register. The combined action is a unitary U. Ideally, starting from $|0\rangle^{\otimes n}$ the output is

$$|\psi_{\text{ideal}}\rangle = U|0\rangle^{\otimes n}. \tag{EST-ideal}$$

In practice, each gate suffers noise. A **coherent error** is a small, deterministic, repeatable unitary mis-rotation—the gate implements $e^{-i\theta_a G_a}U_a$ instead of U_a, where G_a is a Hermitian generator (a Pauli or Pauli string) and $\theta_a \in \mathbb{R}$ is the unknown error angle.

> Coherent errors accumulate *quadratically* in circuit depth: after L layers the logical error is $O(L^2\theta^2)$, while stochastic (random) errors grow only as $O(L\theta^2)$. This makes long circuits extremely sensitive to systematic miscalibration and parasitic ZZ coupling between transmon qubits.

The parameter vector

Collect all K unknown coherent error angles into a vector:

$$\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_K) \in \mathbb{R}^K, \quad \theta_a = \text{error angle at gate location } a. \tag{EST-theta}$$

Goal: estimate $\boldsymbol{\theta}$ from measurement data so it can be corrected.

Randomized Compiling (RC): What It Does

The RC layer

Randomized compiling inserts, around each hard gate U_a, a random single-qubit Pauli P drawn uniformly from $\mathcal{P} = \{I, X, Y, Z\}$. Because $P^2 = I$, the pair (P, P) around the gate leaves the ideal circuit unchanged:

$$P \cdot U_a \cdot P = U_a \quad \text{(logical circuit unchanged)}. \tag{EST-RC-identity}$$

But the coherent error $e^{-i\theta_a G_a}$ transforms under conjugation:

$$P \cdot e^{-i\theta_a G_a} \cdot P = e^{-i\theta_a (PG_aP)} = \begin{cases} e^{-i\theta_a G_a} & \text{if } [P, G_a] = 0 \\ e^{+i\theta_a G_a} & \text{if } \{P, G_a\} = 0 \end{cases}. \tag{EST-sign-flip}$$

For each gate location a, define the **sign label** $s_a \in \{+1, -1\}$: it is $+1$ if the chosen Pauli commutes with G_a and -1 if it anticommutes.

The twirl label vector

For a circuit with L layers and one coherent parameter per layer, the full label for one shot is:

$$\mathbf{s} = (s_1, s_2, \ldots, s_L) \in \{+1, -1\}^L. \tag{EST-label-def}$$

The label $\mathbf{s}$ is drawn from a distribution $\pi(\mathbf{s})$ that is **completely independent of the error angles $\boldsymbol{\theta}$**—it is just the result of dice rolls on the classical computer that programs the gates.

The outcome probability

Given a specific label $\mathbf{s}$ and true error angles $\boldsymbol{\theta}$, the circuit's measurement outcome is b with probability:

$$P(b \mid \mathbf{s}, \boldsymbol{\theta}). \tag{EST-P-model}$$

Standard RC **averages over $\mathbf{s}$** and uses only the marginal distribution:

$$\bar{P}(b \mid \boldsymbol{\theta}) = \sum_{\mathbf{s}} \pi(\mathbf{s})\, P(b \mid \mathbf{s}, \boldsymbol{\theta}). \tag{EST-P-marg}$$

> The average over $\mathbf{s}$ converts the effective error channel into a *Pauli channel*: each coherent error appears as an equal mixture of $\pm\theta_a$ rotations. Their first-order contributions cancel in the marginal (EST-P-marg), leaving only a second-order stochastic residual. This converts quadratic coherent accumulation into linear stochastic accumulation.

The prevailing conclusion: because first-order coherent information vanishes in $\bar{P}$, the twirl destroys it.

This work shows the conclusion is wrong.

Fisher Information: The Precision Currency

Fisher information quantifies how much a measurement tells you about an unknown parameter. All bounds below are established textbook results.

Classical Fisher information

For a probability model $P(b \mid \theta)$ (single parameter), the Fisher information is:

$$F(\theta) = \sum_b P(b \mid \theta) \left(\frac{\partial}{\partial \theta} \ln P(b \mid \theta) \right)^2 = -\sum_b P(b \mid \theta) \frac{\partial^2}{\partial \theta^2} \ln P(b \mid \theta). \tag{EST-FI-def}$$

For a binary outcome $b \in \{0, 1\}$ with mean $m(\theta) = P(b = 1|\theta)$:

$$F(\theta) = \frac{(\partial_\theta m)^2}{m(1-m)}. \tag{EST-FI-binary}$$

For a K-parameter model, Fisher information generalises to a $K \times K$ matrix:

$$[\mathcal{I}(\boldsymbol{\theta})]_{ab} = \sum_b P(b \mid \boldsymbol{\theta})\, \partial_a \ln P \cdot \partial_b \ln P. \tag{EST-FI-matrix}$$

The Cramér–Rao bound

Established Fact 1 (Cramér–Rao, 1945–1946)**.** *Any unbiased estimator $\hat{\theta}$ of θ from M i.i.d. shots satisfies:*

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{M \cdot F(\theta)}. \tag{EST-CRB}$$

High Fisher information ⇒ low estimation variance ⇒ fewer shots needed.

Quantum Fisher information

The quantum Fisher information (QFI) is the maximum classical Fisher information over all possible measurements. For a pure state $|\psi(\theta)\rangle = e^{-i\theta G/2}|\psi_0\rangle$:

$$F_Q(\theta) = 4\,\mathrm{Var}_{|\psi_0\rangle}(G/2) = \langle G^2\rangle - \langle G\rangle^2 \quad \text{(generator variance)}. \tag{EST-QFI}$$

The QFI is the per-shot information ceiling. Any measurement achieves $F(\theta) \leq F_Q(\theta)$.

The Fisher chain rule

Established Fact 2 (Fisher information chain rule, classical statistics)**.** *For two jointly observed quantities (b, s) with parameter θ:*

$$F_{b,s}(\theta) = F_b(\theta) + \mathbb{E}_b[F_{s|b}(\theta)], \tag{EST-chain-rule}$$

where $F_{s|b}(\theta) = \sum_s P(s \mid b, \theta)(\partial_\theta \ln P(s \mid b, \theta))^2$. The joint observation is always at least as informative as the marginal.

Three Fisher Matrices for the RC Setup

Define three Fisher information matrices for the RC problem ($K \times K$, compared in the Loewner positive-semidefinite order $\succeq$):

$$\begin{aligned}
&F_{\text{joint}}(\boldsymbol{\theta}) = \text{Fisher info of joint distribution } P(b, \mathbf{s} \mid \boldsymbol{\theta}) \quad \text{[observe both outcome AND label]} \\
&F_{\text{cond}}(\boldsymbol{\theta}) = \mathbb{E}_{\mathbf{s}}\big[\mathcal{I}(\boldsymbol{\theta} \mid \mathbf{s})\big] \quad \text{[average info across all possible labels]} \\
&F_{\text{marg}}(\boldsymbol{\theta}) = \text{Fisher info of } \bar{P}(b \mid \boldsymbol{\theta}) \quad \text{[standard RC: labels discarded]}
\end{aligned} \tag{EST-three-matrices}$$

The central question: how do these three compare?

The Conservation Law (Theorem 1) — New Result

Because the label $\mathbf{s}$ is drawn independently of the error angles $\boldsymbol{\theta}$, observing $\mathbf{s}$ does not tell you about $\boldsymbol{\theta}$ directly—but it changes the probability of each outcome b. The label acts like a "decoder key": once you know it, the outcome becomes much more informative about the error. Standard RC throws away the decoder key.

Step 1: Score identity

The joint log-likelihood is:

$$\ln P(b, \mathbf{s} \mid \boldsymbol{\theta}) = \ln \pi(\mathbf{s}) + \ln P(b \mid \mathbf{s}, \boldsymbol{\theta}).$$

Since $\pi(\mathbf{s})$ is **independent of $\boldsymbol{\theta}$** (the dice rolls don't depend on the errors), its derivative is zero:

$$\partial_a \ln P(b, \mathbf{s} \mid \boldsymbol{\theta}) = \partial_a \ln P(b \mid \mathbf{s}, \boldsymbol{\theta}).$$

Taking the outer product and averaging over $(b, \mathbf{s})$:

$$\boxed{F_{\text{joint}}(\boldsymbol{\theta}) = F_{\text{cond}}(\boldsymbol{\theta})}. \qquad \textbf{(NEW-score-id)}$$

The joint Fisher information equals the average conditional Fisher information. This is exact for any noise, any circuit, any $\boldsymbol{\theta}$.

Step 2: Chain rule decomposition

Factor the joint probability as:

$$P(b, \mathbf{s} \mid \boldsymbol{\theta}) = \bar{P}(b \mid \boldsymbol{\theta}) \cdot P(\mathbf{s} \mid b, \boldsymbol{\theta}).$$

The log-score splits:

$$\partial_a \ln P(b, \mathbf{s}|\boldsymbol{\theta}) = \underbrace{\partial_a \ln \bar{P}(b|\boldsymbol{\theta})}_{\text{from marginal}} + \underbrace{\partial_a \ln P(\mathbf{s}|b, \boldsymbol{\theta})}_{\text{from label given outcome}}.$$

The cross-term in $F_{b,s}$ vanishes because the second score has zero mean: $\mathbb{E}_{\mathbf{s}|b}[\partial_a \ln P(\mathbf{s}|b, \boldsymbol{\theta})] = \partial_a \sum_{\mathbf{s}} P(\mathbf{s}|b, \boldsymbol{\theta}) = 0$. Therefore:

$$F_{\text{joint}} = F_{\text{marg}} + \boldsymbol{\Delta}, \qquad \boldsymbol{\Delta} \equiv \mathbb{E}_b[F_{\mathbf{s}|b}] \succeq 0. \qquad \textbf{(NEW-chain-rc)}$$

$\boldsymbol{\Delta}$ is the average Fisher information about $\boldsymbol{\theta}$ that the label carries *given* the outcome. It is positive-semidefinite (a Gram matrix of scores, averaged).

Step 3: The conservation law

Combining **(NEW-score-id)** and **(NEW-chain-rc)**:

$$\boxed{F_{\text{cond}}(\boldsymbol{\theta}) = F_{\text{marg}}(\boldsymbol{\theta}) + \boldsymbol{\Delta}(\boldsymbol{\theta}) \succeq F_{\text{marg}}(\boldsymbol{\theta}).} \qquad \textbf{(NEW-conserve)}$$

- F_{marg}: the Fisher information standard RC uses (marginal over labels).
- $\boldsymbol{\Delta}$: the Fisher information RC throws away (in the label-outcome correlation).
- Their sum F_{cond} is conserved—it equals the average optimal-measurement information, regardless of how thoroughly the twirl scrambles the errors.
- Standard RC sets labels to zero. This work recovers $\boldsymbol{\Delta}$ by retaining them.

The complementarity formula

For a single parameter θ with mean twirl sign $\mu = \mathbb{E}[s] \in [-1, +1]$ and tailoring strength $T = 1 - |\mu|$:

$$F_{\text{marg}}(\theta) = \frac{\mu^2 \cos^2\theta}{1 - \mu^2 \sin^2\theta} \xrightarrow{\theta \to 0} \mu^2 F_0 = (1-T)^2 F_0. \qquad \text{(EST-Fmarg-formula)}$$

$$\underbrace{F_{\text{marg}}}_{\text{RC keeps}} = (1-T)^2 F_0, \qquad \underbrace{\boldsymbol{\Delta}}_{\text{RC discards}} = (2T - T^2) F_0, \qquad F_{\text{marg}} + \boldsymbol{\Delta} = F_0.$$

(NEW-complementarity)

At full twirl $T = 1$: $F_{\text{marg}} = 0$ (RC is completely blind to coherent error) but $\boldsymbol{\Delta} = F_0 = F_Q$ (the full quantum limit sits intact in the discarded labels).

Identifiability Matrix — New Result

When the circuit has K error parameters and the twirl applies one random Pauli per gate site, labels at *different sites* may be correlated if a single Pauli gate simultaneously dresses two error generators.

Define the **label correlation matrix**:

$$\Sigma_{ab} = \mathbb{E}[s_a s_b] \in [-1, +1], \quad a, b \in \{1, \ldots, K\}. \qquad \textbf{(NEW-Sigma)}$$

With $\Sigma = I$ (independent labels), $F_{\text{cond}} = \mathcal{F}$ (diagonal in the error basis). In general:

$$\boxed{F_{\text{cond}} = \mathcal{F} \circ \Sigma}, \qquad \textbf{(NEW-Fcond-Sigma)}$$

where $\mathcal{F}_{ab} = \sum_b g_a(b) g_b(b) / P_0(b)$ is the Fisher sensitivity matrix, $g_a = \partial_{\theta_a} P|_{\theta=0}$, and $\circ$ denotes the Hadamard (entry-wise) product.

Identifiability rank = $\text{rank}(\mathcal{F} \circ \Sigma)$. Choosing the twirl group controls Σ—and hence which error parameter *combinations* can be separately recovered. Decorrelating labels for pairs (a, b) whose errors must be distinguished maximises $\text{rank}(\mathcal{F} \circ \Sigma)$. This is **experiment design at zero circuit cost**.

PTM Decomposition Lemma (Theorem 2, Part A) — New Result

The Pauli transfer matrix

Any single-qubit quantum channel $\mathcal{E}$ can be written in the ordered Pauli basis $\{I/\sqrt{2}, X/\sqrt{2}, Y/\sqrt{2}, Z/\sqrt{2}\}$ as a 4×4 real matrix (the PTM):

$$\Lambda = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{t} & A \end{pmatrix}, \tag{EST-PTM}$$

where $\mathbf{t} = (t_X, t_Y, t_Z)^\top$ is the **non-unitality vector** (zero iff $\mathcal{E}$ is unital, e.g. for depolarizing) and A is a 3×3 contraction matrix.

Effect of Pauli twirling on the PTM

Established Fact 3 (Gambetta et al. 2012; Nielsen 2002)**.** *Twirling over the Pauli group $\{I, X, Y, Z\}$ diagonalises the PTM:*

$$\Lambda_{\text{tw}} = \text{diag}(1, \lambda_X, \lambda_Y, \lambda_Z), \quad \lambda_Y = A_{YY}. \tag{EST-twirl-PTM}$$

All off-diagonal elements of A are set to zero. The diagonal entries $(\lambda_X, \lambda_Y, \lambda_Z)$ are the Pauli fidelities*, measurable by cycle benchmarking.*

The decomposition lemma

Lemma 1 (PTM decomposition under twirl — new)**.** *For any CPTP channel $\mathcal{E}$ with PTM as in* (EST-PTM)*, the Pauli-twirled adjoint channel acting on the Y quadrature satisfies:*

$$\mathcal{E}^\dagger_{\text{tw}}(Y) = t_Y\, I + \lambda_Y\, Y, \qquad \lambda_Y = A_{YY}, \tag{NEW-PTM-lemma}$$

where $t_Y = \text{tr}[\mathcal{E}(I) \cdot Y]/2$ is the non-unitality projection onto the Y axis.

Proof. Twirling diagonalises Λ (Fact **??**). The I-row is $(1, 0, 0, 0)$ always (trace preservation), so the I-component maps to itself under any twirl. The Y-column picks out $\lambda_Y = A_{YY}$ after twirling kills off-diagonal terms. The residual t_Y is the surviving non-unitality component on the Y axis. □

Amplitude damping example. Channel: $K_0 = \text{diag}(1, \sqrt{1-p_1})$, $K_1 = \sqrt{p_1}|0\rangle\langle 1|$, rate $p_1 = 1 - e^{-\Delta t/T_1}$.

$$\mathcal{E}(Y) = K_0 Y K_0^\dagger + K_1 Y K_1^\dagger = \sqrt{1-p_1}\, Y - \tfrac{ip_1}{2}(I + Z). \tag{EST-AD-compute}$$

Reading off:

$$\lambda_Y = \sqrt{1-p_1} = e^{-\Delta t/(2T_1)}, \qquad t_Y = -\tfrac{ip_1}{2}. \tag{EST-AD-result}$$

The non-unitality offset t_Y is a fixed function of the physical damping rate p_1—it does **not** depend on the twirl label s.

Noise-Aware Unbiased Recovery (Theorem 2, Part B) — New Result

Effect of noise on the label-outcome signal

Applying Lemma 1, incoherent noise gives:

$$m(s) \equiv \mathbb{E}[b \mid s, \theta] = v \cdot s \cdot \sin\theta + o, \qquad \textbf{(NEW-noisy-mean)}$$

where $v = \lambda_Y \in [0, 1]$ is the **visibility** and $o = t_Y$ is the label-independent offset.

Four consequences

A. Offset cancels under full twirl. Since $\mathbb{E}[s] = 0$ at full twirl ($T = 1$):

$$\mathbb{E}[s \cdot b] = \mathbb{E}_s[s \cdot m(s)] = v \sin\theta \underbrace{\mathbb{E}[s^2]}_{=1} + o \underbrace{\mathbb{E}[s]}_{=0} = v \sin\theta. \qquad \textbf{(NEW-offset-cancel)}$$

The non-unitality offset o vanishes exactly. Label-independent noise does not bias the label-correlation estimator.

B. Naive estimator is shrink-biased. Ignoring v and inverting directly:

$$\hat{\theta}_{\text{naive}} = \arcsin\big(\mathbb{E}[s \cdot b]\big) = \arcsin(v \sin\theta) \approx v\theta < \theta. \qquad \text{(EST-naive-bias)}$$

The noise shrinks the recovered angle by the factor v.

C. Noise-aware estimator is exactly unbiased. Divide by the visibility before inverting:

$$\boxed{\hat{\theta} = \arcsin\left(\frac{\mathbb{E}[s \cdot b]}{v}\right) = \theta.} \qquad \textbf{(NEW-unbiased-est)}$$

D. Self-consistency: v from the marginal. The visibility $v = \lambda_Y = A_{YY}$ is the YY diagonal entry of the twirled PTM. This is exactly the Pauli fidelity that cycle benchmarking already measures from the *same run's marginal data*.

$$v = \lambda_Y = 1 - 2 \sum_{P:\{P,Y\}=0} c_P \quad \text{(Pauli channel case)}, \quad \text{or} \quad v = A_{YY} \text{ (general)}.$$

(NEW-v-from-marg)

No additional circuits are required.

Incoherent noise acts as a multiplicative shrink v on the label signal, plus an additive offset o. The offset cancels under full twirl. The shrink is measured from the marginal (stochastic sector) data. Dividing by v gives an exactly unbiased coherent-error estimate from data that is already being collected by any group running randomized compiling. The full characterisation pipeline closes within one RC run.

Depth Scaling — New Result

Heisenberg propagation of error generators

For L Trotter cycles, each applying ideal $R_x(\alpha)$ then coherent error $R_z(s_\ell\theta)$, push each per-cycle error to the output frame using the Heisenberg propagation:

$$\tilde{G}_\ell = U_{\ell+1:L}\, G_\ell\, U_{\ell+1:L}^\dagger \quad (\text{where } G_\ell = Z). \tag{EST-heis-prop}$$

For the single-qubit Trotter example:

$$\tilde{G}_\ell = \cos\big((L-\ell)\alpha\big)\, Z - \sin\big((L-\ell)\alpha\big)\, Y. \tag{EST-G-tilde}$$

The pre-error ideal state is $|\chi\rangle = R_x(L\alpha)|0\rangle$, with expectation:

$$\langle \tilde{G}_\ell \rangle_{|\chi\rangle} = \cos(\ell\alpha). \tag{EST-Gl-expect}$$

Labeled Fisher information at depth L

Using independence of signs across cycles ($\mathbb{E}[s_\ell s_{\ell'}] = \delta_{\ell\ell'}$), the total labeled Fisher information is:

$$\boxed{\mathbb{E}_{\mathbf{s}}[F_Q] = \sum_{\ell=1}^{L} \big(1 - \langle \tilde{G}_\ell \rangle^2\big) = \sum_{\ell=1}^{L} \sin^2(\ell\alpha) \;\longrightarrow\; \frac{L}{2} \quad (\text{generic } \alpha).}$$

(NEW-depth-formula)

Comparison to no-twirl coherent accumulation

Without the twirl (all $s_\ell = +1$), coherent accumulation is:

$$F_Q^{\text{coh}}(L) = \left(\sum_{\ell=1}^{L} \sin(\ell\alpha)\right)^2 \le \frac{1}{\sin^2(\alpha/2)} \quad (\text{bounded for generic } \alpha). \tag{EST-coh-accum}$$

And for standard RC (labels discarded):

$$F_{\text{marg}} = O(\theta^2) \approx 0 \quad \text{at all depths.} \tag{EST-marg-zero}$$

- Labeled RC: $\mathbb{E}[F_Q] \sim L/2$ — grows *without bound*.
- Coherent (no twirl): bounded by $1/\sin^2(\alpha/2)$.
- Standard RC (marginal): ≈ 0 at all depths.

On a generic Hamiltonian simulation circuit, the labeled estimator *eventually surpasses* the no-twirl coherent accumulation. This is counter-intuitive: adding randomisation and keeping the labels is better than doing nothing. The sign randomisation converts destructively interfering coherent phases into an additive sum of non-negative variances.

Regime of validity. Equation (**NEW-depth-formula**) requires α bounded away from $2\pi k/n$ for small n. Multi-qubit generalisation ($\tilde{G}_\ell$ become Pauli strings) is verified numerically on the 2-qubit TFIM to machine precision.

The Silent Kernel — New Result

What makes an error invisible?

The response matrix is:

$$A_{ka} = i\langle\phi_a|[G_a, M_k^{(a)}]|\phi_a\rangle, \qquad \textbf{(NEW-A-matrix)}$$

where $|\phi_a\rangle = W_a^\dagger|\psi\rangle$ is the ideal state at the error location and $M_k^{(a)} = W_a^\dagger M_k W_a$ is the probe observable back-propagated to that location.

An error $e^{-i\theta_a G_a}$ is **fundamentally silent** if $A_{ka} = 0$ for all probes M_k. This occurs iff:

$$\tilde{G}_a|\psi\rangle \propto |\psi\rangle, \qquad \textbf{(NEW-silent-cond)}$$

i.e. the *propagated* generator $\tilde{G}_a = W_a G_a W_a^\dagger$ acts as a scalar on the output state. The error then contributes only a global phase: $e^{-i\theta_a \tilde{G}_a}|\psi\rangle = e^{-i\theta_a\lambda}|\psi\rangle$, undetectable.

Bell state example

Output state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$. Its stabiliser group is $\{+XX, +ZZ, -YY, +II\}$. Check:

$$XX|\Phi^+\rangle = |\Phi^+\rangle \quad \text{(eigenvalue } +1,\ \text{stabiliser, silent).} \qquad \text{(EST-XX-stab)}$$

$$ZZ|\Phi^+\rangle = |\Phi^+\rangle \quad \text{(eigenvalue } +1,\ \text{stabiliser, silent).} \qquad \text{(EST-ZZ-stab)}$$

$$YY|\Phi^+\rangle = -|\Phi^+\rangle \quad \text{(eigenvalue } -1,\ \text{destabiliser, still silent).} \qquad \text{(EST-YY-dest)}$$

All three give global phases and are undetectable. The six remaining two-qubit Pauli strings (XY, XZ, YX, YZ, ZX, ZY) are active and recoverable.

For a stabilizer output state $|\psi\rangle$, the fundamentally undetectable coherent errors are exactly those whose propagated generators satisfy (**NEW-silent-cond**). These are the generators that either stabilise ($\lambda = +1$) or destabilise ($\lambda = -1$) the state—both give global phases. The identifiable error subspace has dimension $= 15 - |\text{silent kernel}|$ for two-qubit Pauli generators.

The Three-Step Protocol

Any group running RC can implement the following at zero additional circuit cost:

Step 1. Log labels. During the RC run, record the per-layer Pauli sequence $\mathbf{s} = (s_1, \ldots, s_L)$ per shot as a classical bit string. This is already computed by the classical control system; it just needs to be stored alongside the measurement outcome b.

Step 2. Estimate visibility from the marginal. Standard cycle benchmarking on the same run gives the Pauli fidelities $\lambda_Y = A_{YY}$. This is the visibility v. No new circuits.

Step 3. Recover and correct. For each error direction a, compute:

$$\hat{\theta}_a = \arcsin\left(\frac{\mathbb{E}[s_a \cdot b]}{v_a}\right).$$

Apply a counter-rotation $R(-\hat{\theta}_a)$ at the gate location in the next circuit compilation.

Output: a full coherent-error map of the device, updated after every RC job, unbiased under all standard incoherent channels.

Worked Example: Single Qubit, $L = 2$ Cycles

Worked example

Setup. One qubit, two Trotter cycles, $\alpha = \pi/4$, true error angle $\theta^* = 0.2\,\text{rad}$, no incoherent noise.
Circuit. Cycle 1: apply $R_x(\pi/4)$, then coherent error $R_z(s_1\theta^*)$. Cycle 2: apply $R_x(\pi/4)$, then coherent error $R_z(s_2\theta^*)$. Probe: measure Y (apply $S^\dagger H$ before Z-measurement).
Step 1: compute propagated generators. From (EST-G-tilde) with $\alpha = \pi/4$:

$$\tilde{G}_1 = \cos((2-1)\cdot\tfrac{\pi}{4})\,Z - \sin((2-1)\cdot\tfrac{\pi}{4})\,Y = \tfrac{1}{\sqrt{2}}Z - \tfrac{1}{\sqrt{2}}Y.$$
$$\tilde{G}_2 = \cos((2-2)\cdot\tfrac{\pi}{4})\,Z - \sin((2-2)\cdot\tfrac{\pi}{4})\,Y = Z.$$

Ideal output state: $|\chi\rangle = R_x(\pi/2)|0\rangle$, Bloch vector $(0, -1, 0)$.

$$\langle\tilde{G}_1\rangle = \cos(\tfrac{\pi}{4}) = \tfrac{1}{\sqrt{2}} \approx 0.707, \quad \langle\tilde{G}_2\rangle = \cos(\tfrac{\pi}{2}) = 0.$$

Step 2: labeled Fisher information (from (NEW-depth-formula)).

$$\mathbb{E}_s[F_Q] = \sin^2\!\left(\tfrac{\pi}{4}\right) + \sin^2\!\left(\tfrac{\pi}{2}\right) = \tfrac{1}{2} + 1 = 1.5.$$

Both parameters contribute; cycle 2 contributes maximally (generator is Z, perfectly aligned with the Bloch equatorial plane of $|\chi\rangle$).
Step 3: coherent accumulation for comparison (from (EST-coh-accum)).

$$F_Q^{\text{coh}} = \left(\sin(\tfrac{\pi}{4}) + \sin(\tfrac{\pi}{2})\right)^2 = \left(\tfrac{1}{\sqrt{2}} + 1\right)^2 \approx (1.707)^2 = 2.914.$$

At $L = 2$, coherent (2.914) still beats labeled (1.5). But for generic α, coherent is bounded while labeled grows. By $L \sim 6$ they cross.
Step 4: single-shot label-outcome correlation. Draw sign pattern $\mathbf{s} = (+1, -1)$. Outcome probabilities from $P(b|s,\theta) \approx \frac{1}{2}(1 + b\cdot s\cdot\sin\theta\cdot[\text{propagated weight}])$ give:

$$\mathbb{E}[s\cdot b] = s_1\cdot\langle b_1\rangle + s_2\cdot\langle b_2\rangle \approx v\sin\theta^*\cdot\underbrace{(\langle\tilde{G}_1\rangle + \langle\tilde{G}_2\rangle)}_{0.707+0} = 1.0\cdot 0.199\cdot 0.707 = 0.141.$$

Step 5: recover θ. With $v = 1$ (no noise) and the accumulated correlation:

$$\hat{\theta} = \arcsin\!\left(0.141/0.707\right) = \arcsin(0.199) \approx 0.200\text{ rad} = \theta^*.\ \checkmark$$

Step 6: verify conservation. $F_{\text{cond}} = 1.5$. From the marginal: $F_{\text{marg}} \approx O(\theta^2) \approx 0.04$ (small). Check: $\Delta = 1.5 - 0.04 = 1.46 \approx F_{\text{cond}} - F_{\text{marg}}$. Conservation holds to within $O(\theta^2)$. □

Real-World Applications

Real-world applications

The following applications become available once this work is public and researchers adopt the label-retention modification to their RC pipelines. All require only existing hardware and existing RC circuits, with the single change of storing the per-shot label vector.

1. Free coherent-error diagnostic on every RC run.
Any lab running RC—IBM, Google, IonQ, Quantinuum, RIKEN, ETH—can add coherent-error estimation at zero additional QPU cost. Every circuit already run gives a fresh estimate of $\boldsymbol{\theta}$. The device's coherent drift (ZZ crosstalk, off-axis rotations, pulse miscalibration) is continuously monitored without dedicated calibration experiments.

2. Adaptive transpilation.
The recovered $\boldsymbol{\theta}$ is fed into the compiler. Before submitting the next job, the compiler inserts counter-rotations $R(-\hat{\theta}_a)$ at each error location. Over a session, the cumulative fidelity improvement is essentially free. This closes the loop between error characterisation and error correction without QEC.

3. Coherent-error-aware decoders.
Fault-tolerant systems use syndrome extraction circuits that are already Clifford and already twirled in most implementations. The coherent-error map recovered from the labels tells the decoder where systematic biases are coming from, enabling a *coherent-aware* prior that reduces logical error rates.

4. Device calibration at scale.
On 100+ qubit processors, running GST or RPE for every gate is infeasible. The labeled-RC approach produces a coherent-error map across the full device from workload data—no dedicated calibration circuits, no offline characterisation slots.

5. Complement to Pauli-noise learnability tools.
The stochastic Pauli rates are already extracted from the marginal by cycle benchmarking (Chen et al. 2023). Adding label retention gives the coherent rates from the same shots. The two together constitute a *complete* in-situ noise characterisation: stochastic from the marginal, coherent from the labels.

6. Variational algorithm calibration loops.
VQE and QAOA circuits are run in many repetitions. Each repetition produces an RC dataset. The labeled estimator tracks how the coherent error evolves with gate count and repetition, allowing variational optimisers to de-trend systematic drift from the fidelity landscape.

7. Benchmarking and verification.
When verifying that a hardware vendor's processor matches its specification, the coherent-error map provides a quantitative fingerprint of systematic deviations—more sensitive than aggregate error rates and requiring no extra experiments beyond the verification circuits already being run.

Summary: Established vs. New

Established tool	**How used here**	**Status**
Fisher chain rule	Splits $F_{\text{joint}} = F_{\text{marg}} + \mathbf{\Delta}$	existing
Score independence	Proves $F_{\text{joint}} = F_{\text{cond}}$	existing
QFI $= 4\,\text{Var}(\text{generator})$	Sets the conserved ceiling F_0	existing
PTM formalism	Gives Λ block structure	existing
Pauli twirl diagonalises PTM	Derives $\lambda_Y = A_{YY}$	existing
Randomized compiling (RC)	The protocol being analysed	existing
Heisenberg propagation	Propagates errors to output	existing
Cramér–Rao bound	Lower-bounds estimator variance	existing
New result	**What it gives**	**Status**
Conservation law $F_{\text{cond}} = F_{\text{marg}} + \mathbf{\Delta}$	Proves info is relocated, not destroyed	NEW
Complementarity $(1-T)^2 + (2T - T^2) = 1$	Exact trade vs. twirl strength	NEW
PTM decomposition lemma	Noise splits into shrink + offset	NEW
Noise-aware estimator $\hat{\theta}$	Exactly unbiased recovery	NEW
Self-consistency (v from marginal)	No extra circuits needed	NEW
Identifiability matrix $\mathcal{F} \circ \Sigma$	Design handle on separability	NEW
Depth formula $\sum \sin^2(\ell\alpha) \to L/2$	Labeled beats coherent at depth	NEW
Silent kernel characterisation	Exact identifiable subspace	NEW

All blue equations above are new.
All gray equations are established and used as tools.